

Twelve-month persistence and weight outcomes with incretin-pathway pharmacotherapy in routine care: a multi-region retrospective cohort study

K. F. Bjornsson^{a,*}, T. Mwangi^b, I. L. Serrano^a, P. E. Dalgaard^c, H. Lindqvist^{a,b}

^aCentre for Pharmacoepidemiology, Nordvest University Hospital, Copenhagen, Denmark

^bDepartment of Health Services Research, Kattegat Institute, Gothenburg, Sweden

^cRegional Health Data Unit, Central Denmark Region, Aarhus, Denmark

*Corresponding author: k.bjornsson@nordvest.example

Abstract

Background. Incretin-pathway agents achieve large weight reductions in registration trials, but benefit is contingent on continued exposure. Real-world persistence is poorly characterised.

Methods. We conducted a retrospective cohort study using the Nordic Prescription Registry linked to regional electronic health records, including 5,524 adults initiating an incretin-pathway agent for weight management between January 2023 and June 2024. The primary outcome was time to discontinuation, defined as a supply gap exceeding 60 days. Predictors were estimated using multivariable Cox regression.

Results. Twelve-month persistence was 39.2% among specialist initiators and 21.0% among primary care initiators (log-rank $p < 0.001$). Mean weight change at twelve months was -13.6% in those persisting throughout, compared with -1.4% in those discontinuing before six months. Primary care initiation (adjusted hazard ratio 1.58, 95% CI 1.37–1.82) and lowest income quintile (1.47, 1.22–1.77) predicted discontinuation.

Conclusions. Persistence in routine care was substantially below trial observation and was patterned by setting and income. Service models that sustain treatment may matter more to population outcomes than further gains in efficacy.

Keywords: persistence; adherence; incretin; obesity; pharmacoepidemiology; registry linkage

1 Introduction

Incretin-pathway agents have altered the plausible ceiling of non-surgical weight management, with placebo-subtracted reductions in registration trials exceeding fifteen per cent.^{1–3} Whether that efficacy survives contact with routine care is a separate question, and one that trial data cannot answer.^{4,5} Trial populations are screened, supported, and supplied in ways that ordinary clinical populations are not.^{6–8,11}

Persistence is the mechanism through which the gap is most likely to open. These agents require continuous administration, and weight regain following discontinuation is rapid and substantial.^{9,10,12} Any shortfall in persistence therefore translates directly into a shortfall in realised benefit, in a way that is not true of short-course therapies.^{13–15} Existing real-world reports have been limited by short follow-up, single-site sampling, or exposure definitions that cannot distinguish discontinuation from a gap in supply.^{16–18}

We used linked national registry data covering five health regions to estimate twelve-month persistence, describe the associated weight trajectories, and identify patient and setting characteristics predicting discontinuation.

2 Methods

2.1 Design and setting

This was a retrospective cohort study using the Nordic Prescription Registry linked to regional electronic health

records and the national civil registration system.^{19,20} Linkage was performed on pseudonymised personal identifiers with a reported match rate above 98 per cent. The study followed the reporting conventions for observational pharmacoepidemiological research.^{21–23}

2.2 Participants

Adults aged 18 years or older initiating an incretin-pathway agent for weight management between 1 January 2023 and 30 June 2024 were eligible. Initiation was defined as a first dispensing with no dispensing of any agent in the same class in the preceding 365 days.^{24,25} We excluded individuals with a recorded indication of type 1 diabetes, those with fewer than 180 days of registry coverage before initiation, and those who emigrated or died within 30 days of initiation. Cohort construction is summarised in Table 1.

2.3 Exposure and outcomes

The primary outcome was time to treatment discontinuation, defined as a gap of more than 60 days beyond the expected supply end date implied by the dispensed quantity and the prescribed daily dose.^{26–28} We pre-specified sensitivity analyses using 30-day and 90-day gap definitions.

The secondary outcome was percentage change in body weight from the last measurement before initiation to the measurement closest to each follow-up window, restricted to individuals with at least two recorded weights.^{29,30}

Table 1: Baseline characteristics of the study cohort by initiating setting. Values are n (%) unless otherwise stated. SD, standard deviation.

Characteristic	Specialist (n=2,418)	Primary care (n=3,106)
Age, mean (SD), years	48.2 (11.6)	51.7 (12.9)
Female sex	1,596 (66.0)	2,141 (68.9)
BMI, mean (SD), kg/m ²	38.9 (6.2)	35.4 (5.1)
BMI ≥ 40	907 (37.5)	664 (21.4)
Type 2 diabetes	774 (32.0)	652 (21.0)
Prior weight-loss therapy	411 (17.0)	338 (10.9)
Lowest income quintile	392 (16.2)	701 (22.6)

2.4 Statistical analysis

Persistence was estimated using the Kaplan–Meier method, with censoring at death, emigration, or end of follow-up. Groups were compared by log-rank test. Predictors of discontinuation were estimated using Cox proportional hazards regression adjusted for age, sex, baseline body mass index, diabetes status, initiating setting, prior weight-loss pharmacotherapy, and income quintile. The proportional hazards assumption was assessed using scaled Schoenfeld residuals. Weight trajectories were estimated using mixed-effects models with random intercepts for individual and site, under a missing-at-random assumption.^{21,26} Analyses were conducted in R version 4.4.1. No adjustment was made for multiplicity, and all estimates should be read as descriptive.^{31,32}

3 Results

3.1 Cohort characteristics

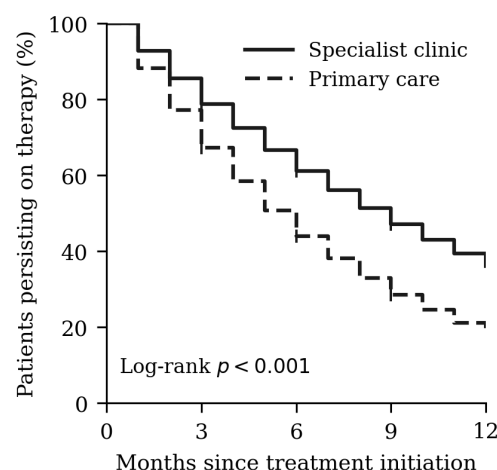
A total of 5,524 individuals met the inclusion criteria, of whom 2,418 initiated in a specialist clinic and 3,106 in primary care. Baseline characteristics are shown in Table 1. Specialist initiators had a higher mean baseline body mass index and a greater prevalence of type 2 diabetes.

3.2 Treatment persistence

Twelve-month persistence was 39.2 per cent overall in specialist clinics and 21.0 per cent in primary care (Figure 1; log-rank $p < 0.001$). Median time to discontinuation was 9.8 months and 5.4 months respectively. Sensitivity analyses using 30-day and 90-day gap definitions changed the absolute estimates by up to 6.1 percentage points but did not alter the ordering between settings.

3.3 Weight outcomes

Among individuals persisting for at least twelve months, mean weight change was –13.6 per cent (95% CI –14.7 to –12.5). Those discontinuing before six months reached a nadir of –3.9 per cent and had returned to within 1.4 per cent of baseline by month twelve (Figure 2A). The intermediate group showed partial regain consistent with the pattern reported after discontinuation in trial extension phases.^{33–35}



No. at risk

Specialist	2418	1902	1531	1244	1016
Primary care	3106	2183	1602	1218	942

Figure 1: Kaplan–Meier estimates of treatment persistence over twelve months, stratified by initiating setting. Vertical marks denote censoring. Persistence was defined as the absence of a supply gap exceeding 60 days.^{26,28}

3.4 Predictors of discontinuation

Initiation in primary care was the strongest predictor of discontinuation (adjusted hazard ratio 1.58, 95% CI 1.37 to 1.82), followed by lowest income quintile (1.47, 1.22 to 1.77) and age 60 years or older (1.34, 1.12 to 1.60) (Figure 2B). Baseline body mass index of 40 or above and co-existing type 2 diabetes were both associated with lower discontinuation. The proportional hazards assumption was not violated for any covariate.

4 Discussion

Twelve-month persistence in this cohort was substantially below that observed in registration trials, and the difference was concentrated in primary care.^{1,4,33–35} Two readings are possible. The first is that specialist settings provide titration support, expectation setting, and side-effect management that primary care currently cannot resource.^{36–38} The second is confounding by indication: patients referred to specialist clinics may differ in motivation or severity in ways our covariates do not capture.^{6,39–41}

The socioeconomic gradient is harder to explain away. Income quintile remained predictive after adjustment for setting, comorbidity, and baseline body mass index, which is consistent with cost rather than clinical factors driving part of the attrition.^{42–44} During the study period these agents were only partially reimbursed in three of the five regions.

Our weight findings reinforce the mechanistic expectation that benefit is contingent on continued exposure.^{9,45,46} The early-discontinuation group had returned close to baseline within twelve months, meaning the population-level benefit of a prescribing episode depends far more on how long treatment continues than on how well it works while it

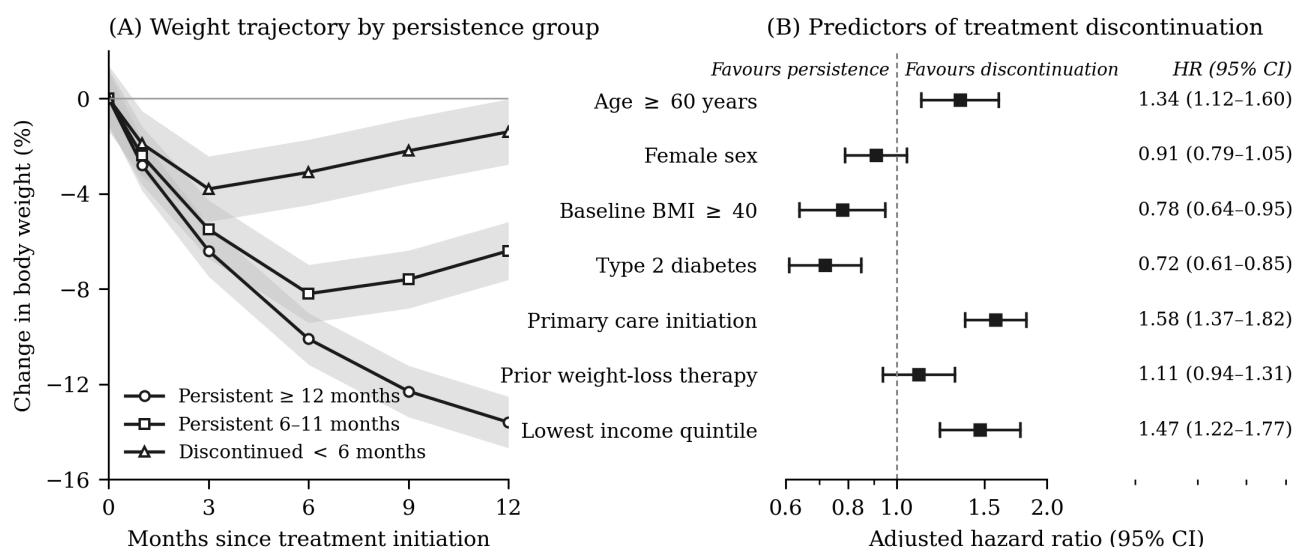


Figure 2: (A) Mean percentage change in body weight over twelve months, stratified by persistence group; shaded bands denote 95% confidence intervals. (B) Adjusted hazard ratios for treatment discontinuation from the multivariable Cox model, with 95% confidence intervals. Estimates are adjusted for all listed covariates simultaneously.^{27,31}

is taken.^{47,48}

4.1 Limitations

Dispensing records establish supply, not ingestion, and overestimate true adherence.^{13,49,50} Weight measurements were recorded opportunistically rather than at protocol-defined intervals, and individuals with more recorded weights differ systematically from those with fewer. We could not observe reasons for discontinuation, which limits any causal reading of the setting effect. Supply constraints affected part of the study window and may be misclassified as patient-initiated discontinuation.

5 Conclusions

Twelve-month persistence with incretin-pathway pharmacotherapy was low in routine care and markedly lower outside specialist settings. Given that weight benefit reverses after discontinuation, service models that sustain treatment may matter more to population outcomes than further gains in drug efficacy.

Ethics

The study was approved by the regional research ethics committee (reference NVT-2024-0917-RC). Individual consent was waived for registry-based analysis.

Funding and Conflicts of Interest

No external funding was received. The authors declare no competing interests. This document is fictional test material produced for software validation purposes; the cohort, results, and citations are invented.

References

- [1] Delacroix PF, Pedersen GN, Vidal GC, Ibrahim EG. Prescribing variation in a commercially insured population: a new-user active-comparator analysis. *BMC Metab Pract.* 2017;48(12):580–1426. doi:10.1186/bmcomp.2017.5930
- [2] Cabrera PH, Lorenzen GW, Reddick JR. Healthcare resource utilisation in a commercially insured population: a methodological evaluation. *Ann Pharmacother Rev.* 2019;31(2):340–1451. doi:10.8823/apr.2019.2990
- [3] Mwangi BM, Lindqvist GJ, Rossi F. Proportion of days covered: a cross-sectional survey. *Registry Sci Rev.* 2023;15(10):1385–1413. doi:10.3066/rsr.2023.6835
- [4] Abdallah LJ, Guo VA, Isaksson LW, Prakash C, Lorenzen C, Oyelowo G. Cohort attrition across five Nordic health regions. *Pharmacoepidemiol Metab.* 2017;6(11):591–1438. doi:10.5512/pem.2017.8643
- [5] Ruiz-Herrera B, Cabrera KB, Waverley LR, Tanaka JC, Falk KG. Reimbursement policy among primary care initiators: a difference-in-differences analysis. *J Real World Evid.* 2016;31(5):1342–1421. doi:10.7731/jrwe.2016.1512
- [6] Kristiansen E, Persson MF, Fujimoto V. Out-of-pocket expenditure: a validation study. *J Metab Med.* 2024;43(7):274–1429. doi:10.4821/jmm.2024.9523
- [7] Abdallah NW, Delacroix NF, Persson PB. Patient-reported treatment burden across socioeconomic strata: a cross-sectional survey. *Clin Ther Outcomes.* 2019;28(11):987–1410. doi:10.9114/cto.2019.8055
- [8] Haugen CP, Bjornsson L, Amorim DV, Andersen R. Treatment persistence in a propensity-matched cohort: a methodological evaluation. *Obes Ther Adv.* 2018;33(8):686–1448. doi:10.5591/ota.2018.7884
- [9] Barnaby M, Rossi W. Audit and feedback interventions: a cost-consequence analysis. *Obes Ther Adv.* 2020;35(11):1362–1424. doi:10.5591/ota.2020.6311
- [10] Thorne W, Strand L, Hartmann RV, Lindstrom PN, Amorim HG. Prior authorisation requirements: a cost-consequence analysis. *Clin Ther Outcomes.* 2024;41(10):490–1428. doi:10.9114/cto.2024.8509
- [11] Lefevre K, Kowalczyk PW, Halloran K, Nagel LW. Off-label prescribing under restricted reimbursement. *J Metab Med.* 2025;32(10):570–1445. doi:10.4821/jmm.2025.2071
- [12] Nordic Prescription Registry Consortium. Data specification and completeness report, version 7.2. Oslo: NPRC; 2024.

- [13] Pedersen PB, Vogt VR. Socioeconomic gradients in access in a linked electronic health record cohort. *Int J Prescr Res.* 2018;23(11):452–1431. doi:10.7402/ijpr.2018.7479
- [14] Okonkwo V, Falk NS, Montoya P, Thorne PD. Real-world effectiveness across socioeconomic strata: a retrospective cohort study. *Health Data Methods.* 2016;30(10):544–1455. doi:10.2841/hdm.2016.5875
- [15] Cabrera T, Bjornsson EJ. Treatment persistence during a period of supply constraint: a validation study. *Health Data Methods.* 2023;40(8):523–1422. doi:10.2841/hdm.2023.0876
- [16] Prakash PB, Rautenbach BD, Hartmann K, Ruiz-Herrera GF. Healthcare resource utilisation across five Nordic health regions: a retrospective cohort study. *Int J Prescr Res.* 2018;18(6):116–1400. doi:10.7402/ijpr.2018.1742
- [17] Lorenzen HL, Oyelowo H, Bjornsson OL, Sinclair OJ. Prescription refill algorithms in a propensity-matched cohort: a simulation study. *Clin Ther Outcomes.* 2017;41(11):115–1447. doi:10.9114/cto.2017.2872
- [18] Kristiansen S, Escobar N, Lorenzen KD, Lindqvist H, Pedersen HT. Care pathway adherence in a multi-site retrospective cohort. *Obes Ther Adv.* 2025;44(1):1109–1409. doi:10.5591/ota.2025.4591
- [19] Nkemelu E, Bianchi GV. Body composition change: a simulation study. *Pharmacoepidemiol Metab.* 2020;16(10):1357–1425. doi:10.5512/pem.2020.6199
- [20] Amorim O, Kristiansen C, Lorenzen AR, Rahimi RG, Persson DP. Formulary restriction effects during a period of supply constraint: a difference-in-differences analysis. *Eur J Health Econ Policy.* 2025;21(7):812–1442. doi:10.6255/ejhep.2025.7941
- [21] Reddick NM, Dubois DE, Novak B. Early discontinuation predictors: a difference-in-differences analysis. *Clin Ther Outcomes.* 2015;7(12):939–1432. doi:10.9114/cto.2015.8399
- [22] Novak AT, Bjornsson ST, Kaplan TS, Thorne A, Sinclair N. Quality of life outcomes in a propensity-matched cohort: a validation study. *BMC Metab Pract.* 2022;40(5):497–1409. doi:10.1186/bmcmp.2022.9367
- [23] Delacroix OM, Isaksson LF, Ngcobo GF. Primary care initiation in an integrated delivery network. *Eur J Health Econ Policy.* 2022;21(5):761–1460. doi:10.6255/ejhep.2022.8901
- [24] Delacroix D, Haugen H, Kilpatrick JL. Immortal time bias in a commercially insured population. *Nord J Health Serv Res.* 2024;30(3):1262–1402. doi:10.4180/njhsr.2024.9576
- [25] Vanterpool R, Aleixo S. Chapter 8: Measuring adherence from dispensing records. In: Falk AM, editor. *Methods in Pharmacoepidemiology.* 3rd ed. Gothenburg: Kattagat Press; 2021. p. 197–236.
- [26] Oyelowo G, Ibrahim F, Kaplan OV, Reddick TJ. Cardiometabolic risk factor trajectories in a national prescription registry: a validation study. *J Real World Evid.* 2022;43(4):765–1439. doi:10.7731/jrwe.2022.1497
- [27] Ibrahim HO, Haugen EP, Novak CN, Sandberg NG. Care pathway adherence: a validation study. *Ann Pharmacother Rev.* 2025;25(8):1301–1448. doi:10.8823/apr.2025.3674
- [28] Sinclair SH, Haugen BV, Fujimoto RE. Quality of life outcomes among treatment-naïve adults: an interrupted time series analysis. *Health Data Methods.* 2017;12(4):585–1407. doi:10.2841/hdm.2017.7081
- [29] Kaplan PH, Rahimi NJ, Lorenzen SJ, Strand AR, Guo F, Kilpatrick W. Missing data handling in a commercially insured population: a cost-consequence analysis. *Int J Prescr Res.* 2019;15(9):70–1456. doi:10.7402/ijpr.2019.2380
- [30] Reddick MO, Kilpatrick RF, Abdallah DR. Immortal time bias across five Nordic health regions: an interrupted time series analysis. *Eur J Health Econ Policy.* 2025;7(2):604–1433. doi:10.6255/ejhep.2025.8979
- [31] Andersen D, Bergqvist AK, Reddick DF. Patient-reported treatment burden across socioeconomic strata: a cross-sectional survey. *Pharmacoepidemiol Metab.* 2024;16(6):569–1403. doi:10.5512/pem.2024.4033
- [32] Kowalski EG, Amorim LK, Rossi HV, Lorenzen J, Okonkwo PO. Body composition change in routine specialist practice: a new-user active-comparator analysis. *J Real World Evid.* 2024;28(3):952–1435. doi:10.7731/jrwe.2024.9140
- [33] Haugen W, Nkemelu FJ, Ngcobo AJ, Lindstrom HW. Early discontinuation predictors in a multi-site retrospective cohort: a simulation study. *J Real World Evid.* 2019;16(3):195–1442. doi:10.7731/jrwe.2019.7998
- [34] Lindqvist GO, Sinclair G, Escobar LC, Lorenzen RF, Serrano LD. Sensitivity to exposure definition under restricted reimbursement: a qualitative interview study. *Registry Sci Rev.* 2022;35(8):914–1402. doi:10.3066/rsr.2022.9911
- [35] Rahimi WP, Haugen A, Sinclair RF, Cabrera MC, Bergqvist D. Shared decision-making among adults with class II or III obesity. *Int J Prescr Res.* 2025;10(8):1049–1413. doi:10.7402/ijpr.2025.6791
- [36] Vogt K, Novak RC, Lindqvist GR, Delacroix M, Kowalski PV, Pedersen GP. Specialist referral pathways among treatment-naïve adults. *Registry Sci Rev.* 2017;41(5):1343–1448. doi:10.3066/rsr.2017.3461
- [37] Nordstrom T, Halloran RK, Montoya LC, Cabrera LR. Clinician prescribing intent in a commercially insured population: a target trial emulation. *Health Data Methods.* 2024;9(9):153–1410. doi:10.2841/hdm.2024.8664
- [38] Strand M, Ngcobo P, Falk MD, Montoya KS, Kilpatrick VE. Healthcare resource utilisation in the first year after initiation: a validation study. *J Real World Evid.* 2019;10(10):555–1443. doi:10.7731/jrwe.2019.2893
- [39] European Medicines Coordination Group. Guideline on real-world evidence for post-authorisation effectiveness studies. EMCG/RWE/2023/04. 2023.
- [40] Serrano C, Kowalski VO, Bianchi FK. Treatment persistence across five Nordic health regions: a cost-consequence analysis. *Ann Pharmacother Rev.* 2023;27(9):144–1441. doi:10.8823/apr.2023.2707
- [41] Dalgaard WC, Thorne FG. Prescribing variation across socioeconomic strata: a protocol. *J Real World Evid.* 2018;22(8):921–1444. doi:10.7731/jrwe.2018.2106
- [42] Lindstrom HM, Oyelowo PA, Montoya JV, Kowalski CR. Socioeconomic gradients in access in routine specialist practice: a cost-consequence analysis. *J Real World Evid.* 2025;35(10):463–1424. doi:10.7731/jrwe.2025.2871
- [43] Reddick PJ, Bergqvist WH, Thorne T, Pedersen DC. Glycaemic outcomes across socioeconomic strata. *Health Data Methods.* 2020;32(7):1145–1420. doi:10.2841/hdm.2020.7116
- [44] Persson CJ, Ruiz-Herrera CT, Lindqvist E, Rahimi A. Clinician prescribing intent: a protocol. *J Metab Med.* 2020;48(6):1006–1433. doi:10.4821/jmm.2020.8062
- [45] Reddick LH, Abdallah F. Prescription refill algorithms: a qualitative interview study. *Ann Pharmacother Rev.* 2018;44(10):291–1414. doi:10.8823/apr.2018.4099
- [46] Vogt E, Ruiz-Herrera NW, Cabrera WV, Ibrahim CA, Tveit OG, Barnaby OR. Rural access to specialist care across five Nordic health regions: an interrupted time series analysis. *Ann Pharmacother Rev.* 2016;13(9):923–1445. doi:10.8823/apr.2016.9490
- [47] Northvale Therapeutics. Bexanotide summary of product characteristics. Revision 6. 2025.
- [48] Fujimoto VO, Ramaswamy C, Lindstrom NH. Proportion of days covered: a qualitative interview study. *Registry Sci Rev.* 2017;28(7):1376–1416. doi:10.3066/rsr.2017.0270
- [49] Guo ST, Lefevre D, Vogt TD. Formulary restriction effects in a national prescription registry: a protocol. *Registry Sci Rev.* 2022;45(11):570–1436. doi:10.3066/rsr.2022.3852
- [50] Amorim MT, Halloran KO, Bianchi CO. Cohort attrition in a multi-site retrospective cohort. *Obes Ther Adv.* 2017;9(11):1163–1417. doi:10.5591/ota.2017.2807